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Research paper

Fast method for calibrated self-discharge measurement of lithium-ion batteries including temperature effects and comparison to modelling

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A B S T R A C T

The self-discharge rate is an important parameter to assess the quality of lithium-ion batteries (LIBs). This paper presents an accurate, efficient, and comprehensive method for measuring and understanding the self-discharge behaviour of LiB cells, considering factors such as temperature and cell to cell variability, as well as underlying electrochemical mechanisms. A method for precise potentiostatic self-discharge measurement (SDM) is demonstrated that is validated by measuring 21 commercial cylindrical 4.7 Ah cells at a state of charge (SoC) of 30%. The self-discharge current ranges between 3 and 6 μA at 23 °C, with an experimental noise level of 0.25 μA. At higher temperatures of 40 °C the self-discharge current increases to 97 μA. The temperature coefficient of voltage (TCV) is experimentally obtained by exposing the cells to a temperature profile with positive and negative step polarities and following the open circuit voltage (OCV) response. Observed TCVs range from +180 μV/K at 40% SoC to −320 μV/K at 0% SoC. For SDM temperature experiments, the cells were set to an SoC with a minimum TCV. From the SDM currents at different temperatures the Arrhenius kinetics and the electrochemical activation energy barrier is determined as 0.94 ± 0.14 eV, indicating chemical side reactions as source of self-discharge. For SDM modelling the electrochemical processes are coupled with a 3D temperature finite element model (FEM) and an electric circuit model resulting in a good overlap with the dynamics and time-constants of the experimental self-discharge curves. The primary challenges addressed are accurately measuring microampere (μA) discharge currents of high-quality cells, reducing measurement time, understanding the temperature dependence of self-discharge, determining activation energy, and demonstrating the applicability and generalization of SDM.

1. Introduction

Lithium-ion batteries (LiBs) are the dominant electrochemical storage technology used in electric vehicles due to their high energy and power densities, as well as their long cycle life (Li et al., 2023). However, LiBs gradually self-discharge over time, which depends on temperature and state of charge (SoC). Self-discharge rates for high-quality LiB cells are low, typically showing a $<$ 2% drop in stored capacity over a month, corresponding to a self-discharge current of $<$ 130 μA for a cell with a capacity of 4.7 Ah. Cells with manufacturing defects or significant degradation can exhibit significantly higher self-discharge rates, eventually leading to thermal runaway (Gao et al., 2019). In the case of a micro internal short-circuit, the self-discharge current can reach up to 1A (Kong et al., 2020; Seo et al., 2020). Varying self-discharge rates between cells in a battery pack can result in voltage imbalances between the cells and a shorter battery pack life (Zheng et al., 2020).

Self-discharge rates vary depending on the cell chemistry, capacity, electrode geometry, electrolyte formulation, impurities, and temperature. The primary cause of self-discharge in LiBs is chemical reactions between the active material and the electrolyte during shelf storage (Sinha et al., 2011). Recent studies show that the progressive growth of the solid electrolyte interphase (SEI) contributes to self-discharge due to the consumption of intercalated lithium in the anode (Yazami and Reynier, 2002), especially at high temperatures (Holzapfel et al., 2004).

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During the thermal storage of LiB cells, lithium intercalated between graphene layers diffuses toward the external edges of the graphene, driven by its reactivity with the electrolyte and impurities, resulting in self-discharge and capacity drop (Yazami and Reynier, 2002). Other factors contributing to self-discharge include cracks in electrodes causing current leakage paths, particle contaminants and dendrite growth producing internal micro-shorts, electrolyte degradation, and electrode passivation by decomposition and film growth. Increased attention is given to the relationship between self-discharge and temperature, particularly in automotive applications where there are large temperature fluctuations (Genieser et al., 2018; Palacín and de Guibert, 2016). A recent study shows that the capacity loss of a battery is greater under high cell stress factors such as high temperature or high SoC (Redondo-Iglesias et al., 2018).

Self-discharge rates vary depending on the cell chemistry, capacity, electrode geometry, electrolyte formulation and impurities, and on temperature. In LiBs, self-discharge is caused primarily by chemical reactions between the active material and the electrolyte during shelf storage (Mushtaq et al., 2022; Shan et al., 2022; Sinha et al., 2011). Recent studies show that the progressive growth of the solid electrolyte interphase (SEI) contributes to the self-discharge due to the consumption of intercalated lithium in the anode (Yazami and Reynier, 2002), especially at high temperatures (Tang et al., 2019). During thermal storage of LiB cells, lithium intercalated between graphene layers diffuses towards the graphene external edges, driven by its reactivity with the electrolyte and impurities resulting in self-discharge and capacity drop (Yazami and Reynier, 2002). Other factors contributing to self-discharge include cracks in electrodes causing current leakage paths, particle contaminants and dendrite growth producing internal micro-shorts, electrolyte degradation and electrode passivation by decomposition and film growth. Increased attention is given to the relationship between self-discharge and temperature, particularly in automotive applications where there are large temperature fluctuations (Palacín and de Guibert, 2016; Utsunomiya et al., 2011). Recent studies show that the capacity loss of a battery is greater under high cell stress factors such as high temperature or high SoC (Guo et al., 2023; Redondo-Iglesias et al., 2018; Teliz et al., 2022).

Methods currently employed to measure the self-discharge of LiBs include open circuit voltage (OCV) decay, which correlates the OCV directly with the SoC of a cell (Ozawa et al., 2003; Ryu et al., 2006; Wang et al., 2002; Zimmerman, 2004). This technique involves monitoring the OCV over several days and estimating the self-discharge current from the rate of OCV change multiplied by the effective capacitance of the cell at the SoC where the OCV was measured. The typical duration for the OCV decay method ranges from 3 to 5 days for classifying cells in manufacturing (Ananda et al., 2022), and up to two weeks for a comprehensive self-discharge characterization in an R&D environment. An alternative to the OCV decay method is measuring the amount of charge required to return to a specific SoC. This method entails charging a cell to a certain SoC using a standard charging rate, allowing the cell to rest for a few weeks, and subsequently discharging the cell to determine the lost charge during rest (Knap et al., 2016; Liao et al., 2022). Here, we introduce a rapid potentiostatic method for directly measuring the self-discharge current, providing precise self-discharge currents within a few hours with a high resolution of 0.25 μA. We demonstrate that the self-discharge measurement (SDM) method is a potent tool capable of measuring the low self-discharge currents of high-quality cells in the range of a few μA. We experimentally investigate how SDM changes with temperature and SoC. The SDM experiments reveal that cell behavior is significantly influenced by temperature and SoC. These measurements are compared to an extended finite element model (FEM) that incorporates electrochemistry, a 3D temperature distribution, and an electric circuit model of the SDM. The FEM model accounts for self-discharge in a cell as chemical side reactions, and the model simulations align remarkably well with the SDM results. In summary, this paper introduces a new method for swiftly and accurately measuring self-discharge currents. The method's precision and efficiency facilitate the evaluation of self-discharge currents even in high-quality cells. The experimental investigation of self-discharge behaviour across different temperatures and SoC levels underscores their notable influence, providing valuable insights into internal processes. Furthermore, the method's accuracy is validated through comparison with an inclusive simulation model, considering factors such as electrochemistry and temperature.

2. Materials and methods

2.1. Battery cells

Self-discharge measurements (SDM) were performed on commercial cylindrical Li-ion 21,700 cells, with a $LiNi_{0.8}Mn_{0.1}Co_{0.1}O_2$ (NMC 811) cathode and a Graphite-SiO_x anode. The cells have nominal capacities of 4.7 Ah, with a nominal voltage of 3.63 V and 17 Wh nominal energy. The cells each have a diameter of 21.1 mm and a height of 70.1 mm, with an energy density of 267 Wh/kg. All cells have undergone several months of shelf storage but no cycling. For SDM, the cells are mounted in a 6 × 6 cell fixture and placed in a temperature-controlled chamber.

2.2. SDM setup and calibration

Fig. 1a illustrates an equivalent electric circuit schematic of the setup used to measure self-discharge current in LIBs. The cell voltage (V_{cell}), with an effective capacitance (C_{eff}) in the range of kF to MF, decreases due to the self-discharge current (I_{SD}) flowing through the parallel self-discharge resistor (R_{SD}) in the range of kΩ to MΩ. In this model, the voltage across the capacitor decreases exponentially over time (Han et al., 2018; Yousif et al., 2017). In the potentiostatic SDM approach, the initial step involves precise measurement of the cell voltage (V_{cell}) and adjustment of the internal voltage source in the SDM setup to match the LIB with a precision of a few micro-volts. The internal voltage source is engineered to minimize drift, staying below 10 μV over 24 h. This voltage source is connected to the LiB through an adjustable output resistance (Rout), typically ranging from 0.1 to 10 Ω. The resulting current (I_{Meas}) is measured by a micro-ammeter with an accuracy of 0.25 μA. The current supplied by the voltage source converges exponentially with a time constant (τ) to an equilibrium value corresponding to the measured self-discharge current (I_{SD}). I_{SD} values are estimated by averaging the last 100 sample points of the SDM curves, using a sampling rate of 1 Hz. Additionally, an exponential fit was employed to validate the estimated I_{SD} values, and both methods show agreement within a 5% accuracy margin. The time constant (τ) depends on the cell's effective capacity (C_{eff}) and the total series resistances, computed as $\tau = (R_{out} + R_{ser}) \bullet C_{eff}$. Consequently, a lower output resistance (R_{out}) leads to quicker convergence and a faster measurement of the self-discharge current. However, it also amplifies undesired artifacts and noise contributions due to the cell's voltage fluctuations. Such voltage fluctuations, caused by factors like temperature changes, are directly transformed into current fluctuations with a gain proportional to $1/R_{out}$. An in-situ calibration is conducted to correct the voltage drop attributed to wire resistance in SDM. To achieve this, the LIB cells are connected to the test fixture, and the wire resistance of each SDM channel is measured using a low current of 1 mA, followed by a 15-minute resting period. Fig. 1b shows an example of raw and calibrated SDM data from two selected cells. A shift in the corrected data toward lower time constants is observed due to the calibrated wire resistances. However, at saturation, the raw and calibrated data align, differing by less than 1 μA in the I_{SD} current, as such, only resulting in reduced measurement time. For temperature evaluation, two sets of self-discharge experiments are conducted. This involves SDM at a constant temperature (23 °C) and SoC (30%) for 21 cells, as well as SDM at five distinct temperatures, including 15 °C, 20 °C, 25 °C, 30 °C, and 40 °C, at 70% SoC. The cells are allowed to equilibrate inside the chamber over a 24-hour period prior to

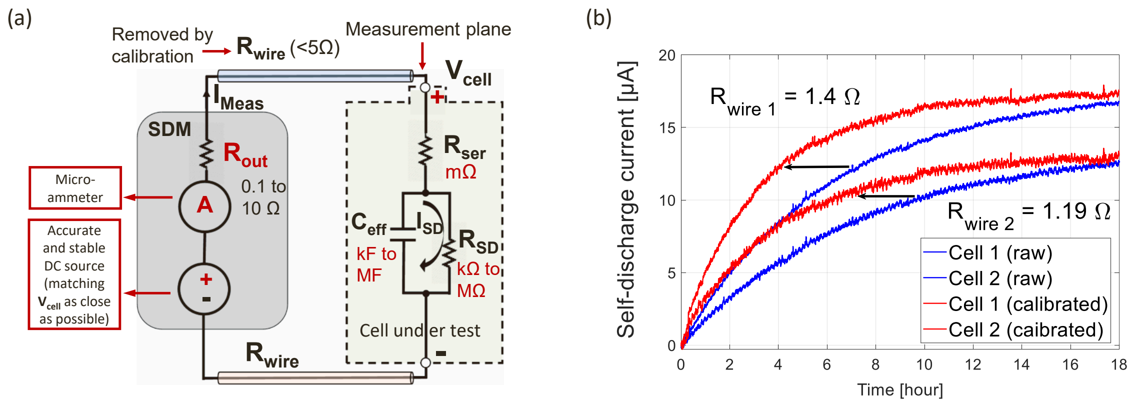

Fig. 1.. SDM concept and calibration. (a) A stable DC source is combined with a sensitive micro-ammeter measuring the current I_{Meas} across the output resistor R_{out}. The LIB is represented by the cell internal series resistance R_{ser}, the effective cell capacitance C_{eff}, and a parallel resistor R_{SD} representing self-discharge. The cell open circuit voltage is denoted by V_{cell} and the internal self-discharge current by I_{SD}. The wire resistance R_{wire} connecting the SDM setup to the LIB is effectively removed in a correction process. (b) Calibration and correction of R_{wire}. Comparison of raw (blue) and corrected (red) self-discharge currents measured with two cells (NMC8711, 21700) at 70% SoC and at 20 °C.

SDM.

2.3. SDM temperature modelling

A finite element model (FEM) based on porous electrode theory (1D Newman model) is utilized to describe the electrochemical processes of the LIB and the temperature-dependent side reactions linked to self-discharge (Fig. 2, left part). In this model, the output is the current, while the inputs are the voltage and temperature. The FEM model incorporates the physics, geometry, and a mesh of the LIB cells employed in the SDM experiments. A 3D temperature model accounting for temperature distribution within the cell and the flow of heat in and out is integrated with the electrochemical 1D Newman model. The principal equations of the 1D model describe the non-isothermal charge/discharge processes. This includes the lithium material balance in the electrolyte, the double-layer capacitance at electrode materials, the electrode film resistance, the ionic concentration, the particle radius of the active material, the exchange current in the electrolyte, and the superficial current density of the cell. Fig. 2 shows the 1D electrochemical model, which considers an NMC811 cathode with 103 µm thickness, a graphite (Li_xC_6 MCMB) anode with 135 µm thickness, a polymer electrolyte (1.2 M $LiPF_6$ in EC: EMC solvent (3:7 by weight)), and a separator of type Celgard 2325 with 15 µm thickness. These physical parameters were verified from extracted battery electrode samples using atomic force microscopy (Al-Zubaidi R-Smith et al., 2021). The 1D Newman model captures temperature influences on the electrode side reactions that lead to a change in the self-discharge current. Also considered are parasitic side reactions of the solvent that lead to an increase in the SEI film thickness on the anode, resulting in an increase in the voltage drop and the self-discharge current as the SEI resistance increases (Palacín and de Guibert, 2016). Additionally, due to the side reactions, the electrolyte volume fraction available for electrolyte charge transport is reduced. The thickness of the SEI film S_{SEI} is calculated from the formed SEI concentration C_{SEI}, as follows:

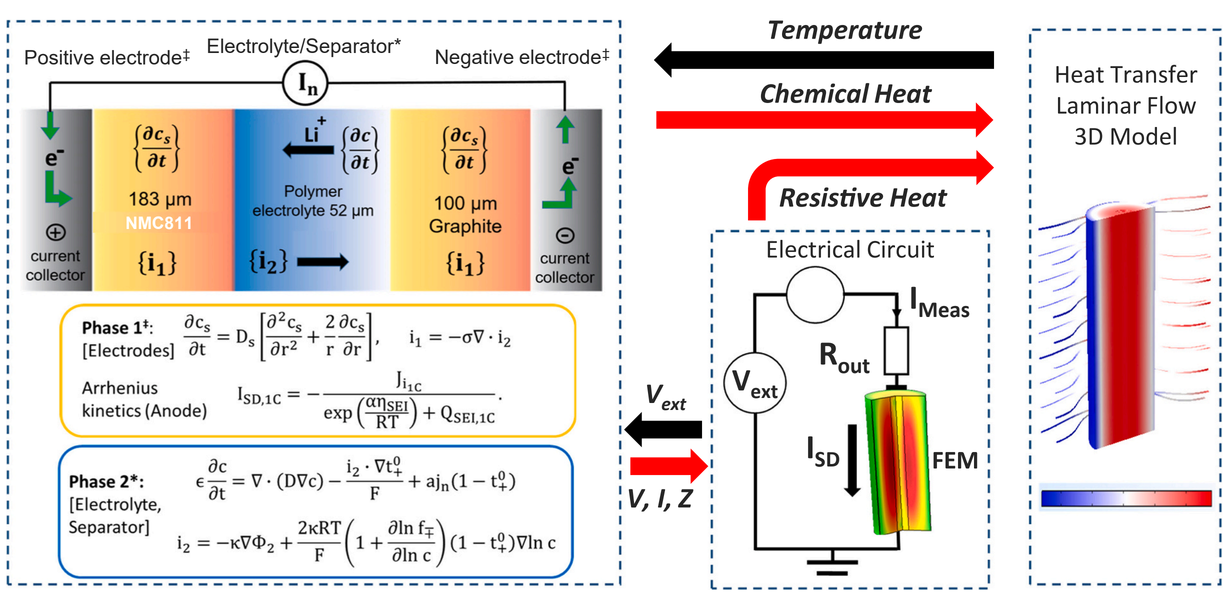

Fig. 2. Combined electrochemical, temperature, and electric circuitry modelling of SDM. Left: The 1D Newman model describes the electrochemistry of the anode, cathode, and electrolyte/separator at the electrode interface, and the self-discharge Arrhenius kinetics at the anode. Middle: Self-discharge electrical circuit model (ECM) combined with the 1D Newman model. Right: Heat generated from electrochemical reactions in the 1D Newman model are coupled with a 3D FEM temperature model, including laminar flow cooling. The relations between the 1D Newman model, the self-discharge, and the 3D temperature model are illustrated by arrows with the physical parameters. Heat, voltage, and current are outputs of the 1D Newman model, and temperature and voltage are the inputs.

$$S_{SEI} = \frac{C_{SEI} \bullet M_p}{A \bullet \rho_p} + S_0 \quad (1)$$

where M_p is the molar weight and ρ_p is the density of the product formed by the side reaction considered as a mixture of organic and inorganic Li-based compounds, and A is the electrode surface area (Ramadass et al., 2004). The initial film thickness S_0 is 1 nm. The change of C_{SEI} is accounted for by solving an additional equation:

$$\frac{\partial C_{SEI}}{\partial t} = \frac{\nu \quad I_{SD,1c}}{nF} \quad (2)$$

where υ is the stoichiometric coefficient, and F is Faraday's constant. The self-discharge current I_{SD} is expressed by an Arrhenius kinetics (Fig. 2, left part),

$$I_{SD} = -\frac{J \bullet i_{1C}}{\exp\left(\frac{\alpha E_{act}}{RT}\right) + Q_{SEI,1C}} \quad (3)$$

Where J is the exchange current density of the side reaction, E_{act} is the activation energy and α the reaction transfer coefficient, and T is the temperature. Q_{SEI} is given as:

$$Q_{SEI} = \frac{C_{SEI}F \quad J \quad f}{Ai_{1c}} \quad (4)$$

with f [1/s] as a fitting parameter related to the SEI properties and i_{1c} is the local reference current density. By substituting Eq. (4) into Eq. (3), the self-discharge current I_{SD} is related to the SEI concertation C_{SEI} as follows:

$$\lim_{T\to\infty} 1/I_{SD} = \ln\frac{C_{SEI}F \quad f}{Ai_{1c} \quad 2} \quad (5)$$

Where C_{SEI} reflects changes of the SEI over time for instance during cell aging. In the LiB cell, heat generation is calculated by coupling two modules in Comsol Multiphysics. First, the heat generated is calculated in the 1D FEM model, based on electrochemical reactions occurring in the anode, cathode, and separator along one dimension. This calculation includes contributions from chemical reaction heat Q_{Chem}, reversible heat generation rate Q_R, and ohmic heat generation rate Q_{Ohm}.

$$Q_{Chem} = FAJ(\varphi_1 - \varphi_2 - U) \quad (6)$$

$$Q_R = FAJT\frac{\partial U}{\partial T} \quad (7)$$

$$Q_{Ohm} = \sigma\left(\frac{\partial\varphi_1}{\partial x}\right)^2 + \alpha\left(\frac{\partial\varphi_2}{\partial x}\right)^2 + \frac{2\alpha RT}{F}(1-t^+)\frac{\partial(\ln c)}{\partial x}\frac{\partial\varphi_2}{\partial x} \quad (8)$$

Here, σ represents the effective electrical conductivity, U denotes the open-circuit potential, t^+ is the transference number of lithium ions, φ_1 and φ_2 represent the potentials in the solid and electrolyte phases, respectively, and x represents the spatial coordinate along one dimension. A comprehensive description of the heat model along with the complete equations can be found in (Cai and White, 2011).

Second, the Heat Transfer in Solids module is employed to simulate heat conduction within the entire LiB cell in three dimensions. In the radial direction of the cell, the thermal conductivity k_{rad} is determined using the following formula:

$$k_{rad} = \frac{\sum L_i}{\sum L_i/k_{L,i}} \quad (9)$$

Where L_i represents the thickness of the various cell layers, and $k_{L,i}$ denotes the thermal conductivities of the respective materials in each layer. The heat capacity Cp_{cell} of the active battery material is determined as follows:

$$Cp_{cell} = \frac{\sum L_i\rho_i Cp_i}{\sum L_i\rho_i} \quad (10)$$

Where ρ_i represents the density of each layer of the cell, and Cp_i denotes the heat capacity of the material of each layer of the cell. The electrochemical reactions are temperature-dependent, and the temperature distribution within the cell evolves due to the heat generated by chemical reactions. These two aspects are coupled by utilizing the average temperature value (T_{Cell}) obtained from the 3D model, and the average heat generation (Q_{Avg}) obtained from the 1D model. All FEM modelling was carried out with COMSOL Multiphysics 6 using the battery toolbox. The main model parameters considered for these simulations are shown in Table 1.

3. Results and discussion

3.1. SDM of 21 cells

The SDM setup measures the current supplied to a cell to maintain its voltage at a constant level, and at equilibrium, the SDM current corresponds to the cell's internal self-discharge current (R-Smith et al., 2020). The SDM current approaches equilibrium with a time constant ranging from minutes to hours, dependent on the cell's effective capacitance and resistance, as well as the effective resistance setting (R_{out}) of the SDM setup. Fig. 3 shows the self-discharge current of 21 LIB cells measured in a 15-hour procedure. The cells were placed in a 6 × 6 fixture (Fig. 3a) and connected to the SDM setup. Prior to SDM, the cells had been stored for a few months and were set to a state-of-charge (SoC) of 30%. The 6 × 6 fixture with the cells was positioned within a temperature chamber set at 23 °C, exhibiting low temperature fluctuations of less than 0.2 K over a 10-hour span. Fig. 3b shows the calibrated self-discharge current of the 21 cells, with I_{Meas} exponentially converging toward the self-discharge value (I_{SD}). I_{SD} is estimated by averaging the last 100 sample points and is validated using an

Table 1
FEM battery model parameters.

Model parameter	Type/Value	Model parameter	Type/ Value
Cathode active material	*$LiNi_{0.8}Mn_{0.1}Co_{0.1}O_2$ (NMC811)*	Particle diameter anode (Schmidt et al., 2020)	*9.5 [μm]*
Anode active material	*Graphite*	Electrolyte concentration (c) (Al-Zubaidi R-Smith et al., 2021)	*1200 [mol/m³]*
Electrode thickness cath. (Waldmann et al., 2020)	*125 [μm] (double side coated)*	Diffusion coefficient (D) (Charbonneau et al., 2020)	*10^{-6} [cm² s⁻¹]*
Electrode thickness anode (Waldmann et al., 2020)	*126 [μm] (double side coated)*	Molar weight of side reaction (M_p)	*0.16 [kg/mol]*
Separator thickness*	*17 [μm]*	Density of the side reaction product (ρ_p)	*1.6×10^3*
Cell capacity*	*4.7 [Ah]*	Nondimensional SEI fit parameter (f)	*200 [1/s]*
Cell dimensions [mm]	*Height [70] Diameter*	Reaction transfer coefficient (α)	*1*
Electrode surface area (A)*	*1100 [cm²]*	Exchange current side reaction (J)	*$8.4 \bullet 10^{-4}$*
State of charge (SoC)*	*70% SoC*	Stoichiometric coefficient (υ)	*1*
Particle diameter cathode (Xuan et al., 2019)	*10 [μm]*	Local current density (i_{1c})	*1.78 [A/m²]*

* Parameters obtained from LiB pre-characterizations

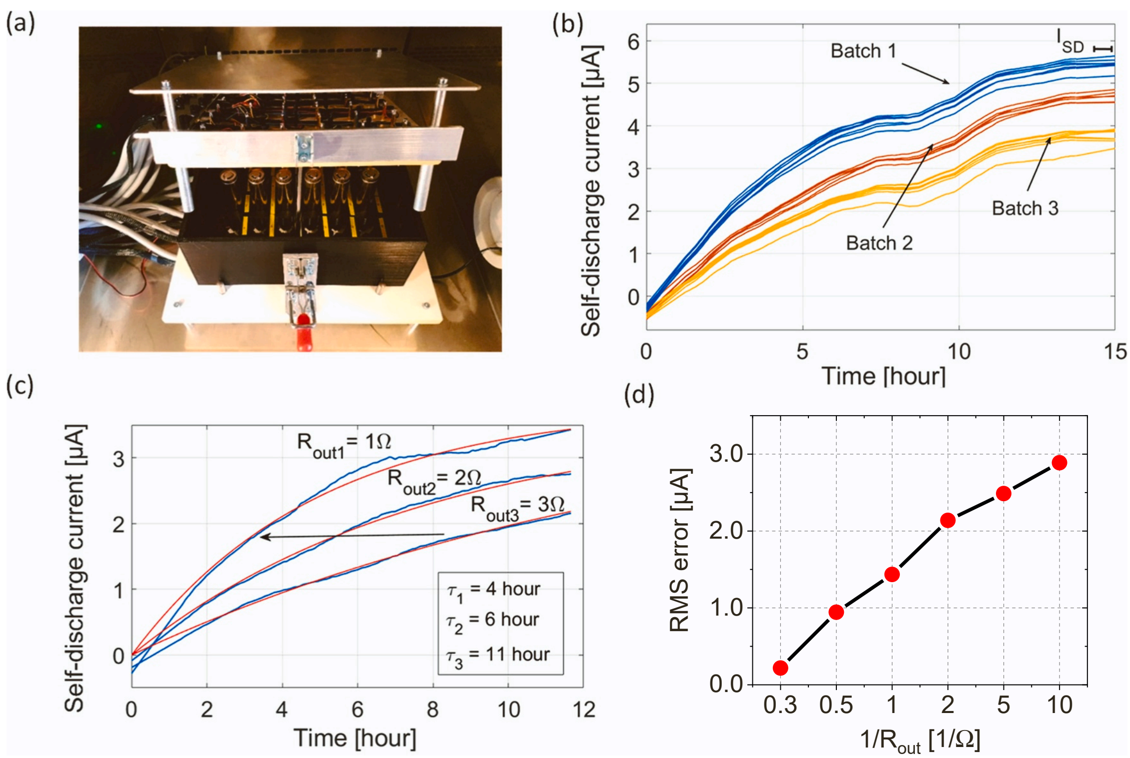

Fig. 3.. SDM results of the 21 cells acquired in the 6 × 6 fixture. a) The SDM fixture for electrically contacting 6 × 6 LIB 21700-NMC cells, mounted in a temperature-controlled chamber. b) Measured current versus time for 21 cells (NMC811, 21700) installed in the 6 × 6 fixture. The cells are measured at 30% SoC and at a temperature of 23 °C with an experiment duration of 15 h. The data range used to estimate the self-discharge current I_{SD} are indicated with a bar. c) Relation between SDM output resistance (R_{out}) and measured current. The reduction in the SDM time constant with lower R_{out} set values is indicated by an arrow. d) The root-mean-square (RMS) error of the current noise as a function of $1/R_{out}$.

exponential fit across the entire time series. It's important to note that when operating within the SDM conditions and within a time frame of less than a day, any changes related to dynamic aging can be considered negligible. For the 21 cells, the self-discharge current (I_{SD}) ranges from 3 μA to 6 μA. Based on the three visible clusters of SDM curves, three batches of cells can be identified, exhibiting minor differences in self-discharge. These differences result from heterogeneous commercial batch sourcing. The corresponding self-discharge resistance (R_{SD}) for the cells is evaluated by the ratio of the OCV and I_{SD}, resulting in R_{SD} = 0.8 ± 0.1 MΩ. A small negative current is observed at the start of the SDM curves, attributed to the practical resolution limit of the measurement hardware of 0.25 μA. Additionally, at intermediate measurement times, a slight dip in the self-discharge current is observed, which is attributed to temperature irregularities during the measurements. Typically, high-quality cells with ∼5 Ah capacity typically exhibit a low self-discharge current of 1–50 μA at room temperature. Conversely, in defective cells with imperfections and leakage currents, self-discharge currents can increase to 150 + μA. To evaluate the effects of altering the output resistance (R_{out}) on the resulting SDM curves, three different Rout values of 1 Ω, 2 Ω, and 3 Ω were employed, and the SDM curves were acquired (Fig. 3c). Lower output resistances (R_{out}) lead to faster equilibration, with the time constant increasing from 4 h at R_{out} = 1 Ω to 11 h at R_{out} = 3 Ω. However, lower output resistance values also result in a reduced signal-to-noise ratio (SNR). Fig. 3d illustrates the root mean square (RMS) error as a function of the inverse of R_{out}, showing a linear increase in the RMS error with lower resistance values. For typical SDM settings with R_{out} = 3 Ω, the RMS error is below 0.3 μA.

3.2. *Temperature effects on the open circuit voltage with respect to SoC*

While the temperature was controlled within a temperature chamber during SDM, any temperature increase or decrease induced changes in the open circuit voltage (OCV), known as the temperature coefficient of voltage (TCV), measured in microvolts per degree Kelvin (μV/K). To determine the TCV, ten cells were set to varying states of charge (SoCs) ranging from 0% to 90% in increments of 10%. A temperature variation profile was applied over several days, encompassing both positive and negative temperature steps. Fig. 4a illustrates the applied temperature profile alongside the monitored OCV response of a selected cell. The TCV is positive when elevated temperature results in an increased OCV. Conversely, the TCV is negative when heightened temperature leads to a reduced OCV, or vice versa. Positive TCV values were observed for cells with SoCs above 30%, while negative TCV values were observed for low SoCs (Fig. 4b). TCVs ranged from +180 μV/K at 40% SoC to −320 μV/K at 0% SoC. At about 20% SoC and 70% SoC a nearly neutral TCV behaviour is observed, where a temperature change causes only a little OCV change. The experimental TCV values were compared to the interpolated TCV values that are used in the Comsol modelling database for the same cell chemistry and cell geometry, and a good agreement is obtained (Fig. 4b, solid line).

3.3. *Temperature dependent SDM curves*

SDM current profiles of twelve cells were experimentally obtained at five different temperatures: 15 °C, 20 °C, 25 °C, 30 °C, and 40 °C, utilizing R_{out} = 2 Ω. For each individual cell, the SDM current profiles at varying temperatures are depicted in Fig. 5a. The self-discharge currents (I_{SD}) increase from 10 μA at 15 °C to 110 μA at 40 °C. A similar self-discharge trend was consistent across all twelve cells, with mean values of 15 μA ± 1.6 μA at 20 °C and 97 μA ± 28 μA at 40 °C (Fig. 5b). Measurements were taken at 70% state of charge (SoC), where the temperature coefficient of voltage (TCV) is minor, and temperature fluctuations minimally affect the open circuit voltage (OCV). The temperature chamber's fluctuations were effectively reduced by wrapping

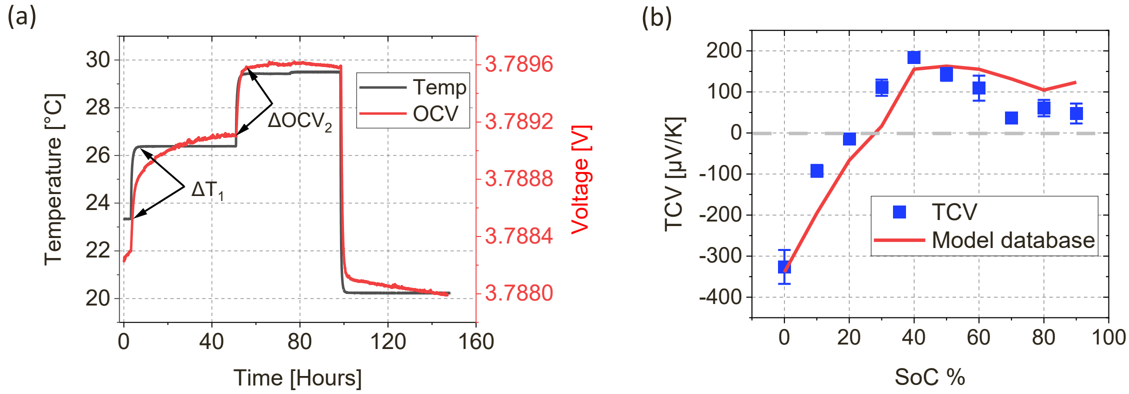

Fig. 4. TCV as a function of SoC. a) The applied temperature profile (black) and the corresponding voltage response (red). Three temperature steps were considered with positive and negative polarities. b) Experimental TCV versus SoC (box plots, blue) compared to the database values used in the Comsol modelling package. The box plot shows the mean value, and the error bars represent the minimum and maximum values obtained from the three temperature steps.

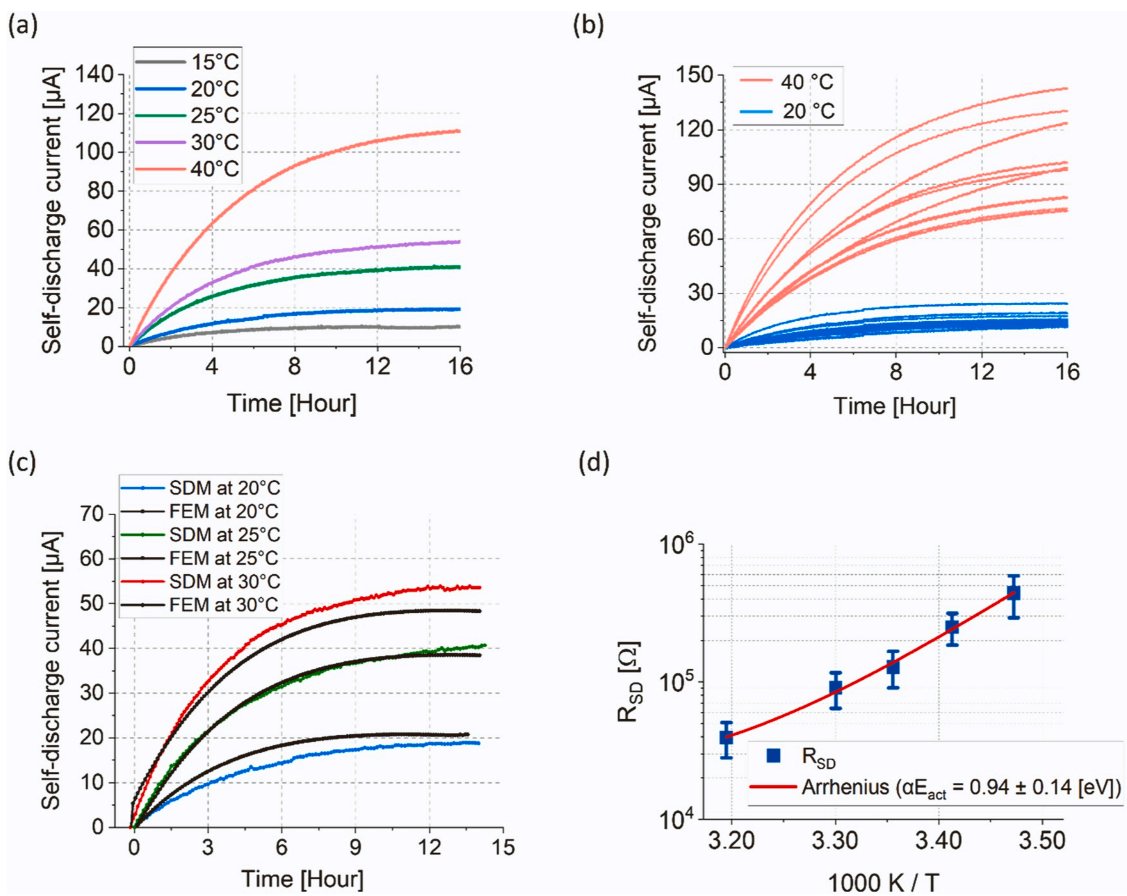

Fig. 5. Temperature dependent SDM curves. a) Self-discharge current profiles of a single cell at 70% SoC measured at five temperatures. b) Self-discharge current of 12 cells at 70% SoC and at two temperatures, 20 °C and 40 °C. c) The measured self-discharge curves are shown for three temperatures (20 °C, 25 °C, and 30 °C) overlaid with the FEM simulation results. c) Arrhenius plot from the SDM data of 12 cells at 5 temperatures.

the cell fixture with thermal insulation and incorporating thermal mass elements around it, acting as a low-pass filter. The amplified self-discharge at higher temperatures is attributed to various mechanisms, including heightened ion migration, accelerated reaction rates, and expedited electrode/electrolyte degradation (Schmidt et al., 2015; Seong et al., 2018; Shan et al., 2022). In Fig. 5b, at elevated temperatures (40 °C, red), the cells display a wider spread of self-discharge current, typically tied to slight variations in the propagation of manufacturing uncertainties, storage, or usage conditions, contributing to differences in self-discharge among batteries with identical chemistry and design (Schmidt et al., 2020; Wildfeuer and Lienkamp, 2021). In essence, the cells employed exhibit commendable self-discharge performance. Notably, even with a self-discharge current of 120 µA/hour, the overall capacity reduction per month remains at approximately 2%. Using FEM simulations, the temperature dependence of the self-discharge current was modelled, including the specific anode and cathode chemistries, the cell capacity, the electronic circuit elements and the 3D temperature distribution within the cell (see Methods section). For the modelling, some cell parameters were measured using microscopy, such as the electrode thickness, and additional parameters were taken from the literature such as the active material particle radius, the electrolyte concentration, and the diffusion coefficient (see Table 1). The modelled self-discharge currents closely align with the experimental SDM values, particularly at intermediate temperatures, exhibiting similar curve shapes and dynamic constants (Fig. 5c). The rise in self-discharge current with increasing temperature is primarily attributed to the heightened electric conductivity of the electrodes (Yazami and Reynier, 2002). Additionally, elevated temperatures lead to an increase in electrolyte conductivity, resulting in a greater current passing through the cell (Alipour et al., 2020), which is not considered in our FEM model. Previous studies have reported comparable findings of increased self-discharge with rising temperature. For instance, in (Schmidt et al., 2015), pouch cells' self-discharge was assessed through discharge capacity tests at two different time points, revealing self-discharge currents ranging from 30 μA to a few hundred μA. Fig. 5d presents the temperature-dependent nature of self-discharge described within an Arrhenius plot using a single activation energy barrier (E_{act}). Self-discharge resistance (R_{SD}) is plotted against the inverse of temperature, with R_{SD} computed by dividing the cell voltage (V_{cell}) by the self-discharge current (I_{SD}). For high-quality cells at room temperature, R_{SD} typically falls within the MΩ range. R_{SD} declines with increased temperatures, resulting in $E_{act} = 0.94 \pm 0.14$ eV. Published studies utilized diverse methods to ascertain the activation energy of similar LIB cells, such as capacity drop tests, yielding similar activation energy values (Alipour et al., 2020; Jow et al., 2018; Schmidt et al., 2015; Spotnitz, 2003). The magnitude of the activation energy implies that chemical side reactions are accountable for the investigated self-discharge, rather than physical defects like internal micro-shorts or current leaks. Generally, changes in Eact indicate variations in the cell defect mechanism and the corresponding cell aging processes (Alipour et al., 2020; Waldmann et al., 2014). The temperature dependence of SDM is relevant for its two main applications. First, when using SDM as a quality gate in cell production, environment temperatures are generally controlled by air conditioning. However, even a fluctuation of a few degrees can change current result by several percents, and this effect must be quantified and considered. Second, in a lab R&D setting, the SDM current response to a defined temperature change stimulus can be treated as a signal yielding information on activation energies, as demonstrated in Fig. 5d.

4. Conclusions

An accurate potentiostatic method for measuring LiB self-discharge was introduced, encompassing hardware calibration and a discussion of temperature effects. The necessary measurement time for self-discharge characterization is significantly reduced to a few hours, compared to the standard open circuit voltage (OCV) method that demands several days of OCV monitoring. In practical terms, the SDM measurement time can be further minimized to as little as 20 min by combining two approaches. First, the output resistance is set to values ranging from 0.1 to 0.2 Ω, effectively shortening the time constant several-fold compared to the output resistance of 1–3 Ω used in this study. Second, data is fitted using the established exponential equation for self-discharge current equilibration, enabling the extrapolation of steady-state current from partial data. The equivalent self-discharge resistance (RSD) at 30% state of charge (SoC) and 23 °C was estimated to be around 0.8 MΩ for the high-quality cells used in this study. Self-discharge currents of 3–6 μA were measured across 21 high-quality cells at room temperature, revealing minor differences of approximately 1 μA between cell types due to the heterogeneous batch sourcing. Similarly, alternative techniques, such as electrochemical impedance spectroscopy with micro-Ohm resolution, can also identify consistent inter-cell variability during the production of high-quality LiB cells (Kasper et al., 2022; Rumpf et al., 2017). The relatively swift measurement time facilitated an exploration of temperature-dependent self-discharge while avoiding changes in SoC and cell aging. Self-discharge currents escalated from 10 μA at 15 °C to 110 μA at 40 °C, allowing for the determination of activation energy from the Arrhenius plot, resulting in Eact = 0.94 eV. The elevated activation energy suggests that chemical side-reactions and mass transport effects are accountable for self-discharge, rather than physical defects or micro-shorts in cells, as the latter display only minor temperature dependence. Temperature coefficient of voltage (TCV) measurements were conducted to assess the impact of temperature on cell OCV. Through exposure to positive and negative temperature profiles, TCV values were determined to be +180 μV/K at 40% SoC and −320 μV/K at 0% SoC, displaying nearly neutral TCV behavior at 70% and 20% SoC. FEM electrochemical modelling was performed including materials properties and cell geometry parameters, resulting in self-discharge current profiles that fit the experimental profiles at intermediate temperatures. The presented SDM method is adaptable for use with LiB cells of different chemical compositions and capacities, functioning under varying ambient conditions to characterize self-discharge spanning from a few μA up to 100 + μA.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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